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ESD protection for integrated circuit devices

Abstract

An integrated circuit device having insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure has been disclosed. The integrated circuit device may include electrostatic discharge (ESD) protection circuit structures. The ESD protection circuit structures may be formed in regions other than the region that the IGFETs are formed as well as in the region that the IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure are formed. By forming ESD protection circuit structures in regions below the IGFETs, an older process technology may be used and device size may be decreased. Furthermore, planar IGFETs of FinFETs may be formed in other regions to decrease device size and improve costs.

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Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 17/030,694 filed on Sep. 24, 2020, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/991,157, filed Mar. 18, 2020, the contents all of which are incorporated by reference herein.

TECHNICAL FIELD

(1) The present invention relates generally to an integrated circuit (IC) device, and more particularly to improving electrostatic discharge (ESD) protection for an IC device.

BACKGROUND OF THE INVENTION

(2) As transistor sizes get smaller, electrostatic discharge (ESD) can be more problematic due to smaller gate dielectric thicknesses and shorter transistor channels. Furthermore, ESD protection circuit structures may be incompatible with new technology and/or consume too much of the active footprint of an integrated circuit device.

(3) In light of the above, it would be desirable to provide ESD protection circuit structures having current discharge capabilities being integrated with new device technology and having a smaller footprint effect on an integrated circuit device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a circuit schematic diagram of an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment.
- (2) FIG. 2 is a circuit schematic diagram of an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment.
- (3) FIGS. 3A and 3B are circuit schematic diagrams of complementary IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure according to an embodiment.
- (4) FIG. 4 is a top plan view of an integrated circuit device including n-type and p-type IGFETS according to an embodiment.
- (5) FIG. 5 is a cross sectional view of integrated circuit device according to an embodiment.
- (6) FIG. 6 is a cross sectional view of integrated circuit device according to an embodiment.
- (7) FIG. 7 is a schematic diagram of an integrated circuit device according to an embodiment.
- (8) FIG. 8 is a schematic diagram of an integrated circuit device according to an embodiment.
- (9) FIG. 9 is a circuit schematic diagram of an ESD protection circuit structure according to an embodiment.
- (10) FIG. 10 is a circuit schematic diagram of an ESD protection circuit structure according to an embodiment.
- (11) FIG. 11 is a circuit schematic diagram of an ESD protection circuit structure according to an embodiment.
- (12) FIG. 12 is a current-voltage diagram of an ESD protection circuit structure according to an embodiment.
- (13) FIG. 13 is a circuit schematic diagram of an integrated circuit device according to an embodiment.
- (14) FIG. 14 is a circuit schematic diagram of an integrated circuit device according to an embodiment.
- (15) FIG. 15 is a circuit schematic diagram of an integrated circuit device according to an embodiment.
- (16) FIG. 16 is a circuit schematic diagram of an integrated circuit device according to an embodiment.
- (17) FIG. 17 is a schematic diagram of an integrated circuit device according to an embodiment.
- (18) FIG. 18 is a schematic diagram of an integrated circuit device according to an embodiment.
- (19) FIG. 19 is a cross-sectional schematic diagram of a planar IGFET that can be formed in a region according to an embodiment.
- (20) FIGS. 20A and 20B are cross-sectional schematic diagrams of a FinFET that can be formed in a region according to an embodiment.
- (21) FIG. 21 is a schematic diagram of an integrated circuit device having an ESD protection circuit structure having a plurality of horizontally current carrying regions that can be vertically aligned above a substrate according to an embodiment.
- (22) FIG. 22 is a diagram of an integrated circuit device according to an embodiment.
- (23) FIG. 23 is a circuit schematic diagram of an internal circuit and an ESD protection circuit structure according to an embodiment.
- (24) FIG. 24 is a circuit schematic diagram of an integrated circuit device including a circuit having

an ESD protection circuit structure according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(25) According to the embodiments set forth below, an input circuit, output circuit, and or input and output circuit including a transistor having a plurality of vertically stacked channels with improved gate control can be electrically connected to an external terminal and have an ESD (electrostatic discharge) circuit having at least portion of the ESD protection circuit structure formed in a layer/region below the input/output circuit. In this way, the footprint of the ESD structure as well as current leakage may be reduced and reliable ESD may be maintained.

(26) Referring now to FIG. 1, an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment is set forth in a schematic diagram and given the general reference character **100**.

(27) The circuit formed on integrated circuit device **100** can include pads (**110**, **120**, and **130**), internal circuit **140**, interface circuit **150**, and ESD protection circuit structures (**160** and **170**). Integrated circuit device **100** may be a semiconductor device.

(28) Pad **110** may receive an externally provided supply potential (for example VDD). Pad **120** may receive an externally provided power supply potential (for example, VSS i.e. ground potential).

(29) Pad **110** may be electrically connected to provide an externally provided supply potential (for example VDD) to internal circuit **140**, interface circuit **150**, and ESD protection circuit structures (**160** and **170**). Pad **120** may be electrically connected to provide an externally provided power supply potential (for example VSS) to internal circuit **140**, interface circuit **150**, and ESD structures (**160** and **170**). Pad **130** may provide and/or receive an external signal (for example, a data or control signal) to or from interface circuit **150** through ESD protection circuit structure **170**.

Interface circuit may receive or generate an internal signal at terminal **152**. Interface circuit **150** may be a buffer circuit that buffers the external and internal signals generated or received.

(30) Internal circuit **140** may receive at least one signal at an input terminal **142** and may provide at least one signal at an output terminal **144**. In another example, internal circuit **140** may be an internal voltage regulating circuit that provides an internal power supply potential. The input terminal **142** and the output terminal **144** of internal circuit **140** are not electrically connected to any pad that can receive or provide a signal external to the integrated circuit device **100**.

(31) In one embodiment, internal circuit **140** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Interface circuit **150** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Both internal circuit **140** and interface circuit **150** may include p-type and n-type IGFETs. In one embodiment, ESD protection circuit structures (**160** and **170**) may include electrical components (such as diodes) formed with a plurality of horizontally disposed cathodes and anodes that can be vertically aligned above a substrate. In one embodiment, ESD protection circuit structures (**160** and **170**) may include electrical components (such as diodes, transistors, silicon controlled rectifiers (SCRs) and/or resistors) formed in the substrate. In one embodiment interface circuit **150** may include electrical components (such as IGFETs) formed in the substrate.

(32) When ESD protection circuit structures (**160** and **170**) are formed in a semiconductor substrate of integrated circuit device **100**, a process having larger critical dimensions (i.e. an older process and less expensive) may be used. The semiconductor substrate may then be sent to a state of the art fabrication facility to form the circuit including insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as will be discussed further in the specification.

(33) Referring now to FIG. 2, an integrated circuit device including a circuit having an ESD

protection circuit structure according to an embodiment is set forth in a circuit schematic diagram and given the general reference character **200**. Integrated circuit device **200** may include a first circuit section **202** and a second circuit section **204**. First circuit section **202** may include circuits that only have external connections to a power supply potential and/or a ground (VSS) potential. Second circuit section **204** may include circuits that have external connections to a power supply potential, a ground potential, and/or a pad coupled to provide or receive an external signal, such as a data signal, control signal or a clock signal, as just a few examples.

(34) First circuit section **202** may include an internal circuit **212** and an ESD protection circuit structure **214**. Internal circuit **212** and ESD structure **214** may each be electrically connected to pad (**210** and **216**). Internal circuit **212** may receive an input signal at an input terminal **218** and may provide an output signal at an output terminal **220**. Pad **210** may receive an external power supply potential, such as VDD and pad **216** may receive an external reference potential such as VSS. In other embodiments, internal circuit **212** may be an internal power supply generator and may receive an external power supply potential at pad **210** and may provide an internal power supply potential to be used by internal circuits. The input terminal **218** and the output terminal **220** of internal circuit **212** are not electrically connected to any pad that can receive or provide a signal external to the integrated circuit device **200**.

(35) Second circuit section **204** may include pads (**250**, **264**, and **266**), an ESD protection circuit structure **252**, an interface circuit **254**, and ESD protection circuit structure **256**. Interface circuit **254** may provide or receive an internal signal at terminal **262** and may provide and/or receive an external signal at pad **266** through ESD protection circuit structure **256**. Interface circuit **254** may be electrically connected to pads (**250** and **264**). Pads (**250** and **264**) may respectively receive an external power supply potential (such as VDD) and a reference potential (such as VSS). ESD structure **252** may be electrically connected between pads (**250** and **264**). ESD structure **256** may be electrically connected to pads (**250**, **264**, and **266**).

(36) In one embodiment, internal circuit **212** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Interface circuit **254** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Both internal circuit **212** and interface circuit **254** may include p-type and n-type IGFETs. In one embodiment, ESD protection circuit structures (**214**, **252**, and **256**) may include electrical components (such as diodes, transistors, and/or resistors) formed with a plurality of horizontally disposed cathodes and anodes that can be vertically aligned above a substrate. In one embodiment, ESD protection circuit structures (**214**, **252**, and **256**) may include electrical components (such as diodes, transistors, SCRs and/or resistors) formed in the substrate. In one embodiment interface circuit **254** may include electrical components (such as IGFETs) formed in the substrate.

(37) When ESD protection circuit structures (**214**, **252**, and **256**) are formed in a semiconductor substrate of integrated circuit device **200**, a process having larger critical dimensions (i.e. an older and cheaper process) may be used. The semiconductor substrate may then be sent to a state of the art fabrication facility to form the circuit including insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as will be discussed further in the instant specification.

(38) Power supply potentials externally provided to pads (**210** and **250**) may be different power supply potentials, such as a first potential (VDD1) for internal circuit **212** and a second potential (VDD2) for interface circuit **254**.

(39) The IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure will now be discussed.

(40) Referring now to FIGS. **3A** and **3B**, circuit schematic diagrams of complementary IGFETs

having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure according to an embodiment are set forth. FIG. 3A is a N-channel (N-type) IGFET **300A** and FIG. 3B is a P-channel (P-type) IGFET **300B**. (41) N-channel IGFET **300A** includes a control gate terminal **310A**, a first source/drain terminal **320A**, and a second source/drain terminal **330A**. Control gate terminal **310A** may be electrically connected to control gate **312A**. Control gate **312A** may be drawn as a plurality of control gates on each side of a plurality of channel region **314A**. In reality, control gate **312A** may surround a plurality of horizontally disposed channel regions **314A** that can be vertically aligned above a substrate.

(42) P-channel IGFET **300B** includes a control gate terminal **310B**, a first source/drain terminal **320B**, and a second source/drain terminal **330B**. Control gate terminal **310B** may be electrically connected to control gate **312B**. Control gate **312B** may be drawn as a plurality of control gates on each side of a plurality of channel region **314B**. In reality, control gate **312B** may surround a plurality of horizontally disposed channel regions **314B** that can be vertically aligned above a substrate.

(43) Referring now to FIG. 4, a top plan view of an integrated circuit device including n-type and p-type IGFETS according to an embodiment is set forth and given the general reference character **400**.

(44) Integrated circuit device **400** may include an N-type IGFET **410A** and a P-type IGFET **410B**.

(45) N-type IGFET **410A** and P-type IGFET **410B** may each include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate.

(46) N-type IGFET **410A** may include drain/source contacts **418A**, a gate contact **416A**, a gate structure **414A**, and vertically aligned and horizontally disposed channel region structures **412A**.

(47) P-type IGFET **410B** may include drain/source contacts **418B**, a gate contact **416B**, a gate structure **414B**, and vertically aligned and horizontally disposed channel region structures **412B**.

(48) Referring now to FIG. 5, a cross sectional view of integrated circuit device **400** is set forth. The cross-sectional view is along the line II-II of FIG. 4.

(49) Integrated circuit device **400** may include a substrate **402**, an insulator layer **422** a N-type IGFET **410A**, and a P-type IGFET **410B**.

(50) N-type IGFET **410A** may include a gate contact **416A**, a gate structure **414A**, and vertically aligned and horizontally disposed channel regions **412A**, and gate insulating layer **420A**. Gate insulating layer **420A** may surround each vertically aligned and horizontally disposed channel regions **412A**.

(51) P-type IGFET **410B** may include a gate contact **416B**, a gate structure **414B**, and vertically aligned and horizontally disposed channel regions **412B**, and gate insulating layer **420B**. Gate insulating layer **420B** may surround each vertically aligned and horizontally disposed channel regions **412B**.

(52) As will be discussed later, IGFETS including vertically aligned and horizontally disposed channel region structures may be used in internal circuits **140** of FIG. 1 or internal circuits **212** of FIG. 2 and/or interface circuit **150** of FIG. 1 and/or interface circuit **254** of FIG. 2, for example and ESD protection circuit structures (**140**, **160**, **170**, **214**, **252**, and/or **256**) may include diodes, transistors, SCRs and/or resistors formed in substrate **402**.

(53) Referring now to FIG. 6, a cross sectional view of integrated device **400** is set forth. The cross-sectional view is along the line I-I of FIG. 4. As shown in FIG. 4, there are two lines I-I as the N-type IGFET **410A** and P-type IGFET **410B** may have similar structures except the materials and/or doping of materials may differ and elements are designated with the suffix "A/B" to illustrate such. Semiconductor device **400** may include a substrate **402**, an insulator layer **422** and N-type and P-type IGFETs (**410A/B**). IGFET **410A/B** may include a gate contact **416A/B**, a gate structure **414A/B**, vertically aligned and horizontally disposed channel regions **412A/B**, gate

insulating layer **420A/B**, and drain/source contacts **418A/B**. Gate structure **416A/B** and gate insulating layer **420A/B** may surround each vertically aligned and horizontally disposed channel regions **412A/B**.

(54) IGFETs (**410A** and **410B**) may be formed by forming a layered crystal of two materials over dielectric region **422**. For example, layers of silicon and silicon germanium may be formed. An etch and deposit step may then be used to form the source/drain regions (**418A** and **418B**). The silicon layer may form the channel regions (**412A** and **412B**). After a vertical etch, the silicon germanium layers may be etched by using a chemical that can selectively etch silicon germanium with the source/drain regions (**418A** and **418B**) used as support structures. Next, the gate dielectric layers (**420A** and **420B**) may be formed using atomic layer deposition, for example of hafnium-dioxide. Then gate structure (**416A** and **416B**) may be formed using atomic layer deposition of a metal layer, for example, tungsten. The n-type IGFETs **410A** may have source/drain regions **418A** doped with n-type carriers, such as phosphorous and/or arsenic, for example. The p-type IGFETs **410B** may have source/drain regions **418B** doped with p-type carriers, such as boron, for example.

(55) As will be discussed later, IGFETS including vertically aligned and horizontally disposed channel region structures may be used in internal circuits **140** of FIG. **1** or internal circuits **212** of FIG. **2** and/or interface circuits **150** of FIG. **1** and/or interface circuit **254** of FIG. **2**, for example and ESD structures (**140**, **160**, **170**, **214**, **252**, and/or **256**) may include diodes, transistors, and/or resistors formed in substrate **402**.

(56) Referring now to FIG. **7**, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **700**.

(57) Integrated circuit device **700** may include similar constituents as integrated circuit device **100** including IGFETs of integrated circuit device **400**, such constituents may be given the same reference character. Integrated device **700** can include ESD protection circuit structures **160** and **170** formed in a substrate **402**, an internal circuit **140**, and an interface circuit **150**.

(58) Integrated circuit device **700** may include different regions. A region **710** may include ESD structures (**160** and **170**) formed in a semiconductor substrate **402**. Another region **720** may include an insulator region **422** which may contain wirings **740**. Wirings **740** may provide an interconnect between ESD structures (**160** and **170**) and interface circuit **150**, internal circuit **140**, and/or pads (**110**, **120**, and **130**). Wirings **740** may be in the form of vertical vias that are formed through insulator layer **422** and/or region **720**. Another region **730** may include internal circuit **140** and interface circuit **150**, as well as wirings **750**, and pads (**110**, **120**, and **130**).

(59) Pad **110** may receive an externally provided supply potential (for example VDD). Pad **120** may receive an externally provided power supply potential (for example VSS), and pad **130** may provide and/or receive an external signal (for example, a data or control signal).

(60) ESD protection circuit structures (**160** and **170**) in region **710** may be formed using planar IGFETs, n-type and/or p-type diffusion regions, and silicon control rectifiers (SCR), for example. Region **720** may include passive elements, such as polysilicon and/or metal resistors, incorporated in ESD protection circuit structures (**160** and **170**).

(61) Internal circuit **140** and interface circuit **150** in region **730** may include p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above substrate **402**. Region **730** may generally have circuitry comprising p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above substrate **402**. Region **710** may include planar IGFETs fabricated using older technologies with more relaxed critical dimensions. In this way, reliable ESD protection circuit structures can be made more cheaply. Another advantage is that the region **730** exclusively has the normal operating circuits (i.e. exclusive of ESD protection circuit structures which only operate when there is an ESD event). For example, if integrated circuit device **700** is a microprocessor, the central processing unit (CPU), bus, and memory would all be located in region **730** and manufactured with a cutting-edge state of the art process having smaller critical dimensions. By forming ESD structure (**160** and **170**) below

the functional circuits, chip size can be reduced.

(62) Referring now to FIG. **8**, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **800**.

(63) Integrated circuit device **800** may include similar constituents as semiconductor device **200** including IGFETs of integrated circuit device **400**, such constituents may be given the same reference character. Integrated circuit device **800** can include ESD protection circuit structures (**214**, **252**, and **256**) formed in a substrate **402**, an internal circuit **212**, and an interface circuit **254**.

(64) Integrated circuit device **800** may include different regions. A region **810** may include ESD protection circuit structures (**214**, **252**, and **256**) formed in a semiconductor substrate **402**. Another region **820** may include an insulator region **422** which may contain wirings **840**. Wirings **840** may provide an interconnect between ESD protection circuit structures (**214**, **252**, and **256**) and interface circuit **254**, internal circuit **212**, and/or pads (**210**, **216**, **250**, **264**, and **266**). Wirings **840** may be in the form of vertical vias that are formed through insulator layer **422** and/or region **820**. Another region **830** may include internal circuit **212** and interface circuit **254**, as well as wirings **850**, and pads (**210**, **216**, **250**, **264**, and **266**).

(65) Internal circuit **212** and ESD protection circuit structure **214** may each be electrically connected to pad (**210** and **216**). Internal circuit **212** may receive an input signal at an input terminal **218** and may provide an output signal at an output terminal **220**. Pad **210** may receive an external power supply potential, such as VDD and pad **216** may receive an external reference potential such as VSS. In other embodiments, internal circuit **212** may be an internal power supply generator and may receive an external power supply potential at pad **216** and may provide an internal power supply potential to be used by internal circuits.

(66) Pad **250** may receive an externally provided supply potential (for example VDD). Pad **264** may receive an externally provided power supply potential (for example VSS), and pad **266** may provide and/or receive an external signal (for example, a data or control signal).

(67) ESD protection circuit structures (**214**, **252**, and **256**) in region **810** may be formed using planar IGFETs, n-type and/or p-type diffusion regions, and silicon control rectifiers (SCR), for example. Region **820** may include passive elements, such as polysilicon and/or metal resistors, incorporated in ESD structures (**214**, **252**, and **256**).

(68) Internal circuit **212** and interface circuit **254** in region **830** may include p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above substrate **402**. Region **830** may generally have circuitry comprising p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above substrate **402**. Region **810** may include planar IGFETs or FinFETs fabricated using older technologies with more relaxed critical dimensions. In this way, reliable ESD protection circuit structures can be made more cheaply. Another advantage is that the region **830** exclusively has the normal operating circuits (i.e. exclusive of ESD structures which only operate when there is an ESD event). For example, if semiconductor device **800** is a microprocessor, the central processing unit (CPU), bus, and memory would all be located in region **830** and manufactured with a cutting edge state of the art process having smaller critical dimensions. By forming ESD protection circuit structures (**214**, **252**, and **256**) below the functional circuits, chip size can be reduced.

(69) FIGS. **9** to **11** illustrate various ESD protection circuit structures that may be formed in regions (**710** and **810**) of FIGS. **7** and **8**.

(70) Referring now to FIG. **9**, a circuit schematic diagram of an ESD protection circuit structure according to an embodiment is set forth and given the general reference character **900**. ESD structure **900** may be a silicon controlled rectifier (SCR).

(71) ESD structure **900** may include bipolar transistors (**Q1** and **Q2**) and resistors (**R910** and **R920**). Bipolar transistor **Q1** may have an emitter terminal connected to a terminal **910**, a base terminal commonly connected to a first terminal of resistor **R920** and a collector terminal of bipolar transistor **Q2**, and a collector terminal commonly connected to a base terminal of bipolar transistor

Q2 and a first terminal of resistor R910. Bipolar transistor Q2 may have an emitter terminal connected to a terminal 920. Resistor R910 may have a second terminal connected to terminal 920. Resistor R920 may have a second terminal connected to terminal 910.

(72) ESD structure 900 may be used as ESD protection circuit structures (160, 170, 214, 252, and/or 256). When used as ESD protection circuit structure (140, 214, or 252) terminal 910 may be electrically connected to pads (110, 210, or 250) respectively, and terminal 920 may be electrically connected to pads (120, 216, or 264) respectively. When ESD protection circuit structure 900 is used as ESD protection circuit structure (170 or 256), terminal 910 may be connected to pads (130 or 266) respectively and terminal 920 may be electrically connected to pads (120 or 264) respectively.

(73) Referring now to FIG. 10, a circuit schematic diagram of an ESD protection circuit structure according to an embodiment is set forth and given the general reference character 1000.

(74) ESD protection circuit structure 1000 can include diodes (D1002 and D1004). Diode D1002 may have a cathode terminal connected to terminal 1020 and an anode terminal connected to terminal 1010. Diode D1004 may have a cathode terminal connected to terminal 1010 and an anode terminal connected to terminal 1030.

(75) ESD protection circuit structure 1000 may be used as ESD protection circuit structures (170 and 256). When used as ESD protection circuit structures (170 or 256), terminal 1020 may be electrically connected to pads (110 or 250), respectively, terminal 1010 may be electrically connected to pads (130 or 266), respectively, and terminal 1030 may be electrically connected to pads (120 or 264), respectively.

(76) Referring now to FIG. 11, a circuit schematic diagram of an ESD protection circuit structure according to an embodiment is set forth and given the general reference character 1100.

(77) ESD structure 1100 can include IGFETs (P1102 and N1102). IGFET P1102 may have a source terminal and gate terminal commonly electrically connected to terminal 1120 and drain terminal electrically connected to terminal 1110. IGFET N1102 may have a source terminal and gate terminal commonly electrically connected to terminal 1130 and drain terminal electrically connected to terminal 1110. IGFET P1102 may be a p-type IGFET and IGFET N1102 may be a n-type IGFET.

(78) ESD structure 1100 may be used as ESD protection circuit structures (170 and 256). When used as ESD protection circuit structures (170 or 256), terminal 1120 may be electrically connected to pads (110 or 250), respectively, terminal 1110 may be electrically connected to pads (130 or 266), respectively, and terminal 1130 may be electrically connected to pads (120 or 264), respectively.

(79) Referring now to FIG. 12, a current-voltage diagram of an ESD protection circuit structure is set forth.

(80) FIG. 12 is a current-voltage diagram of a typical ESD protection circuit structure. For example, FIG. 12 may be a current-voltage (I-V) diagram of SCR 900 of FIG. 9.

(81) For example, the current voltage diagram of FIG. 12 shows the SCR 900 in a forward blocking region 1202 in which there is minimal leakage current, which occurs when there is no ESD event. Once an ESD event occurs and the voltage spikes above a trigger voltage V_{trigger} shown at point 1204 in the I-V diagram of FIG. 12, the SCR 900 snaps back through snap back region 1206 toward a minimum holding voltage V_{holding} at point 1208. Then in the holding region 1210, the SCR functions as a near ideal switch, the slope in holding region 1210 represents the on resistance of the SCR 1200. This slope is proportional to the size of the SCR 1200, thus a larger SCR 1200 dissipates more current at a lower holding voltage in the holding region. In designing the ESD protection circuit structures it is important to place the trigger voltage V_{trigger} at a voltage that will be low enough that the IGFETs formed with vertically aligned and horizontally disposed channel regions in regions (730 and 830) will not breakdown during an ESD event.

(82) Referring now to FIG. 13, a circuit schematic diagram of an integrated circuit device

according to an embodiment is set forth and given the general reference character **1300**.

(83) Integrated circuit device **1300** can include an ESD protection circuit structure **1310** and an input buffer circuit **1320**. ESD protection circuit structure **1310** can be electrically connected to receive an input signal at a pad **1304**. The input signal may pass through the ESD protection circuit structure **1310** to terminal **1314**.

(84) Input buffer circuit **1320** may receive the input signal from terminal **1314** and may provide an output signal at terminal **1308** (terminal **1308** may be an output terminal). Input buffer circuit **1320** may receive an enable signal EN at terminal **1322** and a reference potential Vref at terminal **1324**.

(85) ESD circuit structure **1310** can include an ESD protection circuit structure **1312** and a resistor **R1300**. ESD protection circuit structure **1312** may be electrically connected to pads (**1302**, **1304**, and **1306**). Pad **1302** may receive an externally provided power supply potential, such as VDD. Pad **1306** receive an externally provided power supply potential, such as VSS. Pad **1304** may receive an input signal, such as an address, data, and/or control signal, as just a few examples. ESD protection circuit structure **1312** may be electrically connected to a first terminal of resistor **R1300**. Resistor **R1300** may be electrically connected to terminal **1314**.

(86) ESD structure **1312** may be an ESD protection circuit structure (**900**, **1000**, or **1100**), as just a few examples. In the case of ESD structure (**900**, **1000**, or **1100**), pad **1304** may be electrically connected to terminal (**910**, **1010**, or **1110**), respectively.

(87) Input buffer circuit **1320** may include IGFETs (**P1322**, **P1324**, **N1322**, **N1324**, and **N1326**). IGFET **P1322** may have a source terminal electrically connected to pad **1302** and commonly coupled to a source terminal of IGFET **P1324**. IGFET **P1322** may have a gate terminal and a drain terminal commonly connected to a gate terminal of IGFET **P1324** and a drain terminal of IGFET **N1322**. IGFET **P1324** may have a drain terminal connected to terminal **1308**. IGFET **N1322** may have a gate terminal coupled to receive a signal at terminal **1314** through ESD circuit structure **1310**. IGFET **N1322** may have a source terminal commonly connected to a source terminal of IGFET **N1324** and a drain terminal of IGFET **N1326**. IGFET **N1324** may have a drain terminal connected to terminal **1308** and a gate terminal connected to receive a reference potential Vref at terminal **1324**. IGFET **N1326** may have a gate terminal connected to receive an enable signal EN at terminal **1322** and a source terminal connected to pad **1306**. Input buffer circuit **1320** may operate as a differential input buffer that is enabled when enable signal EN is at a logic high level and disabled when enable signal EN is at a logic low level.

(88) IGFETs (**P1322**, **P1324**, **N1322**, **N1324**, and **N1326**) may each include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. **3A**, **3B**, **4**, **5**, and **6** and may be formed in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively. ESD structure **1312** may be formed in regions **402** in FIGS. **7** and **8**, for example. Resistor **R1300** may be formed in region (**402** and/or **422**) in FIGS. **7** and **8**, for example. Resistor **R1300** may even be formed in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively. Resistor **R1300** may be formed, for example, as a diffusion layer in region **402**, a metal layer in region **422**, and/or a metal layer in region (**702** or **802**) as set forth in FIGS. **7** and **8**, respectively.

(89) Reference potential Vref may provide a threshold voltage for determining the logic level of an input signal received at pad **1304**. For example, if the potential of the input signal received at pad **1304** is greater than reference potential Vref, input buffer circuit **1320** may provide a logic high output at output terminal **1308**. However, if the potential of the input signal received at pad **1304** is less than reference potential Vref, input buffer circuit **1320** may provide a logic low output at output terminal **1308**.

(90) Referring now to FIG. **14**, a circuit schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1400**.

(91) Integrated circuit device **1400** can include an ESD protection circuit structure **1410** and an output buffer circuit **1420**. ESD circuit structure **1410** can be electrically connected to a pad **1404**

where an output signal may be provided externally to the integrated circuit device **1400**. The output signal may pass from output buffer **1420** through the ESD circuit structure **1410** to pad **1404**.

(92) Output buffer circuit **1420** may receive an input signal from terminal **1408** and may provide an output signal at terminal **1414**

(93) ESD circuit structure **1420** can include an ESD structure **1412** and a resistor **R1400**. ESD structure **1412** may be electrically connected to pads (**1402**, **1404**, and **1406**). Pad **1402** may receive an externally provided power supply potential, such as VDD. Pad **1406** receive an externally provided power supply potential, such as VSS. Pad **1404** may receive a signal to be provided externally from integrated circuit device **1400**. ESD protection circuit structure **1412** may be electrically connected to a first terminal of resistor **R1400**. A second terminal of resistor **R1400** may be electrically connected to terminal **1414**.

(94) ESD protection circuit structure **1412** may be an ESD protection circuit structure (**900**, **1000**, or **1100**), as just a few examples. In the case of ESD protection circuit structure (**900**, **1000**, or **1100**), pad **1404** may be electrically connected to terminal (**910**, **1010**, or **1110**), respectively.

(95) Output buffer circuit **1420** may include IGFETs (**P1422** and **N1422**). IGFET **P1422** may have a source terminal electrically connected to pad **1402**. IGFET **P1422** may have a gate terminal and input terminal **1408** and a gate of IGFET **N1422**. IGFET **P1422** may have a drain commonly connected to a drain of IGFET **N1422** and a second terminal of resistor **R1400** at node **1414**. IGFET **N1422** may have a source terminal electrically connected to pad **1406**. Output buffer circuit **1420** may operate as an inverter logic circuit that provides current drive to a signal, such as a data signal or the like that is to be driven to components external to integrated circuit device **1400**.

(96) IGFETs (**P1422** and **N1422**) may each include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. 3A, 3B, 4, 5, and 6 and may be formed in region (**702** or **802**) as set forth in FIGS. 7 and 8, respectively. ESD protection circuit structure **1412** may be formed in regions **402** in FIGS. 7 and 8, for example. Resistor **R1400** may be formed in region (**402** and/or **422**) in FIGS. 7 and 8, for example. Resistor **R1400** may even be formed in region (**702** or **802**) as set forth in FIGS. 7 and 8, respectively. Resistor **R1300** may be formed, for example, as a diffusion layer in region **402**, a metal layer in region **422**, and/or a metal layer in region (**702** or **802**) as set forth in FIGS. 7 and 8, respectively.

(97) Referring now to FIG. 15, a circuit schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1500**. Integrated circuit device **1500** may include like constituents as integrated circuit device **1400** and such constituents may be designated by the same reference character and for brevity will not be discussed. Integrated circuit device **1500** may differ from integrated circuit device **1400** of FIG. 14, in that an output buffer **1520** may have IGFETs (**P1522** and **N1522**) that are formed in substrate **402** and may be planar IGFETs of FinFETs, while other circuits, such as an internal circuit (**140** or **212**) of FIGS. 1 and 2, respectively, may be formed from IGFETs that include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. 3A, 3B, 4, 5, and 6 and may be formed in region (**702** or **802**) as set forth in FIGS. 7 and 8, respectively.

(98) Integrated circuit device **1300** and integrated circuit devices (**1400** and **1500**) may be incorporated into integrated circuit devices (**100** and **200**). Integrated circuit device **1300** may have a separate pad electrically connected to the input terminal **1304** than a pad electrically connected to the output terminal **1404** of integrated circuit devices (**1400** and **1500**). Input buffer circuit **1320** and ESD structure **1310** of integrated circuit device **1300** may be used as interface circuit **150** and ESD circuit structure **170** of integrated circuit device **100** of FIG. 1. Input buffer circuit **1320** and ESD structure **1310** of integrated circuit device **1300** may be used as interface circuit **254** and ESD circuit structure **256** of integrated circuit device **200** of FIG. 2. Output buffer circuit **1420** and ESD structure **1410** of integrated circuit device **1400** may be used as interface circuit **150** and ESD

circuit structure **170** of integrated circuit device **100** of FIG. **1**. Output buffer circuit **1420** and ESD structure **1410** of integrated circuit device **1400** may be used as interface circuit **254** and ESD circuit structure **256** of integrated circuit device **200** of FIG. **2**. Output buffer circuit **1520** and ESD structure **1510** of integrated circuit device **1500** may be used as interface circuit **150** and ESD circuit structure **170** of integrated circuit device **100** of FIG. **1**. Output buffer circuit **1520** and ESD structure **1510** of integrated circuit device **1500** may be used as interface circuit **254** and ESD circuit structure **256** of integrated circuit device **200** of FIG. **2**.

(99) Referring now to FIG. **16**, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1600**.

(100) Integrated circuit device **1600** may be like integrated circuit device **700** of FIG. **7**, except integrated circuit device **1600** may include a resistor **1610** formed in region **730** along with the circuitry comprising p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above semiconductor substrate **402**.

(101) Resistor **1610** may have one terminal electrically connected to pad **130** as well as ESD protection circuit structure **170** and another terminal electrically connected to interface circuit **150**. Resistor **1610** may correspond to resistors **R1400** in integrated circuit devices (**1400** and **1500**) as set forth in FIGS. **14** and **15**. Resistor **1610** may comprise a metal, such as copper, tungsten, aluminum, and/or titanium or even polysilicon, as just a few examples.

(102) Referring now to FIG. **17**, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1700**.

(103) Integrated circuit device **1700** may be like integrated circuit device **700** of FIG. **7**, except integrated circuit device **1700** may include a resistor **1710** formed in region **720** along with wirings **740**.

(104) Resistor **1710** may have one terminal electrically connected to pad **130** as well as ESD protection circuit structure **170** and another terminal electrically connected to interface circuit **150**. Resistor **1710** may correspond to resistors **R1400** in integrated circuit devices (**1400** and **1500**) as set forth in FIGS. **14** and **15**. Resistor **1710** may comprise a metal, such as copper, tungsten, aluminum, and/or titanium or even polysilicon, as just a few examples.

(105) Referring now to FIG. **18**, a schematic diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **1800**.

(106) Integrated circuit device **1800** may be like integrated circuit device **700** of FIG. **7**, except integrated circuit device **1800** may include a resistor **1810** formed in region **710** along with ESD structures (**160** and **170**) and planar IGFETs fabricated using older technologies with more relaxed critical dimensions.

(107) Resistor **1810** may have one terminal electrically connected to pad **130** as well as ESD protection circuit structure **170** and another terminal electrically connected to interface circuit **150**. Resistor **1810** may correspond to resistors **R1400** in integrated circuit devices (**1400** and **1500**) as set forth in FIGS. **14** and **15**. Resistor **1810** may comprise a metal, such as copper, tungsten, aluminum, and/or titanium, or polysilicon or a diffusion layer, as just a few examples.

(108) Resistors (**R1300**, **R1400**, **1610**, **1710**, and/or **1810**) need sufficient resistance to provide a voltage drop between the pad **1404** and the interface circuit **150**. Resistors (**R1300** and **R1400**) may be about 1 k Ω to 10 k Ω .

(109) The process minimum feature size of region **730** may be the control gate length of p-type and n-type IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above semiconductor substrate **402**. Gate length is illustrated with reference to FIG. **6**, in which "L" is the gate length of IGFET **410A/B**. In the embodiments, the minimum gate length may be about 5 nm or less.

(110) The process minimum feature size of region **402** may be substantially greater. For example, a gate length of planar IGFETs formed in region **402** may be 10 nm or greater. An example of a planar IGFET is illustrated in FIG. **19**.

(111) Referring now to FIGS. 19, a cross-sectional schematic diagram of a planar IGFET that can be formed in region 402 is set forth and given the general reference character 1900.

(112) A planar IGFET formed in region 402 can include a semiconductor substrate 1902 in which source/drain regions 1918 may be formed, a gate insulating layer 1920, a control gate 1914 and an insulating layer 1930. Region 402 can include p-type IGFETs and n-type IGFETs. For example, an n-type IGFET may be formed by implanting n-type impurities into source/drain regions 1918 of a p-type semiconductor substrate 1902. A p-type IGFET may be formed by providing a n-type well in semiconductor substrate 1902 and implanting p-type impurities into source/drain regions 1918. Planar IGFET may have a gate length L1 of about 10 nm or greater. In this way, cost may be reduced as compared to the fabrication to IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as in region 702.

(113) The IGFET formed in region 402 can be used as IGFETs (P1522 and N1522) that are formed in substrate 402 as illustrated in FIG. 15, while other circuits, such as an internal circuit (140 or 212) of FIGS. 1 and 2, respectively, may be formed from IGFETs that include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. 3A, 3B, 4, 5, and 6 and may be formed in region (702 or 802) as set forth in FIGS. 7 and 8, respectively.

(114) Referring now to FIGS. 20A and 20B, a cross-sectional schematic diagrams of a Fin field effect transistor (FinFET) type IGFET (i.e. FinFET) that can be formed in region 402 is set forth and given the general reference character 2000.

(115) FIG. 20A may be a cross-sectional schematic diagram of a FinFET along the width of a channel region 2016 and FIG. 20B may be a cross-sectional schematic diagram of a FinFET along the length of a channel region 2016 and between source/drain regions 2018.

(116) A FinFET formed in region 402 can include a semiconductor substrate 2002 in which source/drain regions 2018 may be formed, a gate insulating layer 2020, a control gate 2014 and an insulating layer 2030. Region 402 can include p-type FinFETs and n-type FinFETs. For example, an n-type FinFET may be formed by implanting n-type impurities into source/drain regions 2018 of a p-type semiconductor substrate 2002. A p-type FinFET may be formed by providing a n-type well in semiconductor substrate 2002 and implanting p-type impurities into source/drain regions 2018. FinFET may have a gate length L2 of about 7 nm or greater. In this way, cost may be reduced as compared to the fabrication to IGFETs having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as in region 702.

(117) The FinFET formed in region 402 can be used as IGFETs (P1522 and N1522) that are formed in substrate 402 as illustrated in FIG. 15, while other circuits, such as an internal circuit (140 or 212) of FIGS. 1 and 2, respectively, may be formed from IGFETs that include a control gate that may surround a plurality of horizontally disposed channel regions that can be vertically aligned above a substrate as set forth in FIGS. 3A, 3B, 4, 5, and 6 and may be formed in region (702 or 802) as set forth in FIGS. 7 and 8, respectively.

(118) Referring now to FIG. 21, a schematic diagram of an integrated circuit device having an ESD protection circuit structure having a plurality of horizontally current carrying regions that can be vertically aligned above a substrate is set forth and given the general reference character 2100.

(119) Integrated circuit device includes regions (710, 720, and 730). As noted earlier, region 710 may be formed with a technology node that is older and cheaper than region 730. Region 710 may include planar IGFETs and ESD structures. However, integrated circuit device 2100 may differ in that an ESD structure may be formed in region 730 and may include diodes (D2102 and D2104). Diodes (D2102 and D2104) may include a plurality of horizontally disposed current carrying regions that can be vertically aligned above a substrate region 402. The current carrying regions may include a first impurity doped region 2112 and a second impurity doped region 2114. Each

diode (D**2102** and D**2104**) may include a cathode terminal **2116** and an anode terminal **2118**. The anode terminal **2118** of diode D**2102** may be electrically connected to the cathode terminal **2116** of diode D**2104** and may be electrically connected to a pad **2110** which may be electrically connected to provide or receive an external signal. The cathode terminal **2116** of diode D**2102** may be electrically connected to a pad **2120**. Pad **2120** may receive an externally provided power supply potential, such as VDD. The anode terminal **2118** of diode D**2104** may be electrically connected to a pad **2130**. Pad **2130** may receive an externally provided power supply potential, such as VSS.

(120) The ESD protection circuit structure of FIG. **21** including diodes (D**2102** and D**2104**) may correspond to ESD protection circuit structure **1000** of FIG. **10** and may be used accordingly. Pad **2130** may correspond to terminal **1030**, pad **2110** may correspond to terminal **1010**, and pad **2120** may correspond to terminal **1020**. Likewise, diode D**2102** may correspond to diode D**1002** and diode D**2104** may correspond to diode D**1004**.

(121) Diodes (D**2102** and D**2104**) may be formed by forming a layered crystal of two materials over dielectric region **422**. For example, layers of silicon and silicon germanium may be formed. The silicon layer may form the first and second impurity doped regions (**2112** and **2114**), i.e. the current carrying regions. After a vertical etch, the silicon germanium layers may be etched by using a chemical that can selectively etch silicon germanium with the cathode and anode terminals (**2116** and **2118**) used as support structures. Next, a dielectric layer **2122** (may be formed using atomic layer deposition of a dielectric, for example, silicon dioxide. The first impurity doped region **2112** may be doped with n-type carriers, such as phosphorous and/or arsenic, for example. The second impurity doped region **2114** may be doped with p-type carriers, such as boron, for example. The doping may be done by implantation with a mask layer over regions other than the desired regions to receive the impurities. In this way each of the plurality of horizontally disposed current carrying regions may form a p-n junction diode in parallel with each other.

(122) Diodes (D**2102** and D**2104**) may be formed in conjunction with insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as discussed above.

(123) Referring now to FIG. **22**, a diagram of an integrated circuit device according to an embodiment is set forth and given the general reference character **2100**.

(124) Integrated circuit device **2200** may differ from integrated circuit devices of previous embodiments in that a region **2210** may be disposed between regions (**720** and **730**), otherwise integrated circuit device **2200** may be substantially the same as previous embodiments. Region **2210** may be a crystalline semiconductor layer. For example, region **2210** may be silicon material. Region **2210** may be silicon, silicon carbide, epitaxial silicon, as just a few examples. Region **2210** may improve the manufacturability of layers used to form the horizontally disposed and vertically aligned channel regions. Integrated circuit device **2200** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure formed in region **730** as discussed above, as well as ESD protection circuit structures formed in regions (**710**, **720**, and **730**) as discussed in previous embodiments.

(125) Referring now to FIG. **23**, a circuit schematic diagram of an internal circuit and an ESD protection circuit structure according to an embodiment is set forth and given the general reference character **2300**.

(126) Circuit **2300** can include an internal circuit **2310** and an ESD protection circuit structure **2320**.

(127) Internal circuit **2300** may receive a power supply potential from a pad **2302**. The power supply potential from pad **2302** may be an externally applied power supply potential such as VDD. Internal circuit **2300** may receive a power supply potential from a pad **2306**. The power supply potential from pad **2306** may be an externally supplied power supply potential such as VSS.

Internal circuit may receive an input signal from an input terminal **2308** and provide an output signal at a terminal **2314**. Internal circuit **2310** may include a p-type IGFET **P2312** and an n-type IGFET **N2312**. Both p-type IGFET **P2312** and n-type IGFET **N2312** may each include a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure.

(128) P-type IGFET **P2312** may have a source terminal electrically connected to pad **2302**. N-type IGFET **N2312** may have a source terminal electrically connected to pad **2306**. P-type IGFET **P2312** and N-type IGFET **N2312** may have gate terminals commonly connected to receive the input signal from input terminal **2308** and drain terminals commonly connected to provide the output signal at output terminal **2314**.

(129) Internal circuit **2310** and may be used as internal circuit **140** and/or internal circuit **212** of FIGS. **1** and **2**, respectively. ESD protection circuit structure **2320** may be used as ESD protection circuit structure **160** or any/each or ESD circuit structures (**214** and **252**) of FIGS. **1** and **2**, respectively.

(130) Internal circuit **2310** may not be electrically connected to receive or provide a signal external to the integrated circuit device.

(131) ESD protection circuit structure **2320** may include two ESD protection circuits, a diode **D2324** and an ESD protection circuit **2322**, each electrically connected between pads (**2302** and **2306**). In this way, ESD protection circuit structure **2320** may provide protection for an ESD event at either pad (**2302** or **2306**), that receive externally provided power supply potentials. ESD protection circuit **2322** may be a SCR such as SCR **900** illustrated in FIG. **9**. ESD protection circuit **2322** can be provided in regions (**710** or **810**),

(132) Diode **D2324** can have a cathode terminal electrically connected to pad **2302** and an anode terminal electrically connected to pad **2306**. Diode **D2324** can be formed in regions (**710** or **810**) as a p-n junction or in region (**730** or **830**). When diode **D2324** is formed in regions (**730** or **830**), diode **D2324** may include a plurality of horizontally current carrying regions that can be vertically aligned above a substrate as illustrated with respect to diodes (**D2104** and **D2102**) in FIG. **21**.

(133) Referring now to FIG. **24**, an integrated circuit device including a circuit having an ESD protection circuit structure according to an embodiment is set forth in a circuit schematic diagram and given the general reference character **2400**. Integrated circuit device **2400** may have similar circuit constituents as integrated circuit device **200** of FIG. **2** and such constituents may have the same reference character. Integrated circuit device **2400** may include a first circuit section **202**, a second circuit section **2410**, and a third circuit section **2440**. First circuit section **202** may include circuits that only have external connections to a power supply potential and/or a ground (VSS) potential. Second circuit section **2410** may include circuits that have external connections to a power supply potential, a ground potential, and/or a pad coupled to receive an external signal, such as a data signal, control signal or a clock signal, as just a few examples. Third circuit section **2440** may include circuits that have external connections to a power supply potential, a ground potential, and/or a pad coupled to provide an external signal, such as a data signal, control signal or a clock signal, as just a few examples.

(134) First circuit section **202** may include an internal circuit **212** and an ESD protection circuit structure **214**. Internal circuit **212** and ESD structure **214** may each be electrically connected to pad (**210** and **216**). Internal circuit **212** may receive an input signal at an input terminal **218** and may provide an output signal at an output terminal **220**. Pad **210** may receive an external power supply potential, such as VDD and pad **216** may receive an external reference potential such as VSS. In other embodiments, internal circuit **212** may be an internal power supply generator and may receive an external power supply potential at pad **210** and may provide an internal power supply potential to be used by internal circuits. The input terminal **218** and the output terminal **220** of internal circuit **212** are not electrically connected to any pad that can receive or provide a signal external to the integrated circuit device **200**.

(135) Second circuit section **2410** may include pads (**2412**, **2414**, and **2416**), an ESD protection circuit structure **2422**, an input buffer circuit **2420**, and ESD protection circuit structure **2418**. Input buffer circuit **2420** may receive an external signal at pad **2416** through ESD protection circuit structure **2418** and may provide an internal signal at terminal **2424**. Input buffer circuit **2420** may be electrically connected to pads (**2412** and **2414**). Pads (**2412** and **2414**) may respectively receive an external power supply potential (such as VDD) and a reference potential (such as VSS). ESD structure **2422** may be electrically connected between pads (**2412** and **2414**). ESD structure **2418** may be electrically connected to pads (**2412**, **2414**, and **2416**).

(136) Third circuit section **2440** may include pads (**2442**, **2444**, and **2446**), an ESD protection circuit structure **2448**, an output buffer circuit **2450**, and ESD protection circuit structure **2452**. Output buffer circuit **2450** may receive an internal signal at terminal **2454** and may provide an external signal at pad **2446** through ESD protection circuit structure **2452**. Output buffer circuit **2450** may be electrically connected to pads (**2442** and **2444**). Pads (**2442** and **2444**) may respectively receive an external power supply potential (such as VDD) and a reference potential (such as VSS). ESD structure **2448** may be electrically connected between pads (**2442** and **2444**). ESD structure **2452** may be electrically connected to pads (**2442**, **2444**, and **2446**).

(137) In one embodiment, internal circuit **212** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Input buffer circuit **2420** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Output buffer circuit **2452** may include insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure. Internal circuit **212**, input buffer circuit **2420**, and output buffer circuit **2450** may include p-type and n-type IGFETs. In one embodiment, ESD protection circuit structures (**214**, **2418**, **2422**, **2448**, and **2452**) may include electrical components (such as diodes, transistors, and/or resistors) formed with a plurality of horizontally disposed cathodes and anodes that can be vertically aligned above a substrate. In one embodiment, ESD protection circuit structures (**214**, **2418**, **2422**, **2448**, and **2452**) may include electrical components (such as diodes, transistors, SCRs and/or resistors) formed in the substrate. In one embodiment input buffer circuit **2420** and/or output buffer circuit **2452** may include electrical components (such as IGFETs) formed in the substrate.

(138) When ESD protection circuit structures (**214**, **2418**, **2422**, **2448**, and **2452**) are formed in a semiconductor substrate of integrated circuit device **2400**, a process having larger critical dimensions (i.e. an older and cheaper process) may be used. The semiconductor substrate may then be sent to a state of the art fabrication facility to form the circuit including insulated gate field effect transistors (IGFETs) having a plurality of horizontally disposed channels that can be vertically aligned above a substrate with each channel being surrounded by a gate structure as will be discussed further in the instant specification.

(139) Integrated circuit device **1300** and integrated circuit devices (**1400** and **1500**) may be incorporated into integrated circuit device **2400**. Input buffer circuit **1320** and ESD structure **1310** of integrated circuit device **1300** may be used as input circuit **2420** and ESD circuit structure **2418** of integrated circuit device **2400** of FIG. 24. Output buffer circuit **1420** and ESD structure **1410** of integrated circuit device **1400** may be used as output buffer circuit **2450** and ESD circuit structure **2452** of integrated circuit device **2400** of FIG. 24. Output buffer circuit **1520** and ESD structure **1510** of integrated circuit device **1500** may be used as output buffer circuit **2450** and ESD circuit structure **2452** of integrated circuit device **2400** of FIG. 24.

(140) Power supply potentials externally provided to pads (**210**, **2412**, and **2442**) may be different power supply potentials, such as a first potential (VDD1) for internal circuit **212**, a second potential (VDD2) for input buffer circuit **2420**, and/or a third potential (VDD3) for output buffer circuit

2450.

- (141) Input buffer **1300** of FIG. **13** and output buffers (**1400** and **1500**) of FIGS. **14** and **15** may be incorporated into integrated circuit devices **2400**.
- (142) Integrated circuit devices (**700**, **800**, **1600**, **1700**, **1800**, **2100**, and **2200**) may be contiguous structures, such that, regions may be deposited or bonded in a semiconductor fabrication facility and preferably all formed on a contiguous wafer in a multiple of units and then separated before packaged or set in a multi-chip package. For example, regions (**710**, **720**, and **730**) may be contiguous regions with virtually no separation other than a region border formed by a change of materials. Bonding of regions may be performed using wafer to wafer bonding, for example region **710** may be formed on a first semiconductor wafer and regions (**720** and **730**) may be formed on a second semiconductor wafer, then the first and second wafer may be bonded using a wafer to wafer bonding technique followed by dicing and packaging to form the integrated circuit device. Alternatively, region **710** may be formed on a first semiconductor wafer and regions (**720** and **730**) may be formed on a second semiconductor wafer, then the either the first or second wafer may be diced and a die pick and place may be used to place dies on the first or second intact wafer, followed by dicing and packaging to form the integrated circuit device.
- (143) It is understood that the term pad may be any circuit connection that is electrically connected to provide or receive a signal or a potential externally to the integrated circuit device. Such a connection can be a conduit for an ESD event.
- (144) Electrically connected can be a connection through a wiring other passive component such as a resistor.
- (145) A voltage may be expressed as a potential.
- (146) A signal can be a data or control signal that can transition between logic levels, as just a few examples. A signal is not a power supply potential used to provide power to circuitry.
- (147) Other electrical apparatus other than semiconductor devices may benefit from the invention.
- (148) While various particular embodiments set forth herein have been described in detail, the present invention could be subject to various changes, substitutions, and alterations without departing from the spirit and scope of the invention. Accordingly, the present invention is intended to be limited only as defined by the appended claims.

Claims

1. An integrated circuit device, comprising: a first region; a second region formed over the first region; an internal circuit coupled between a first power supply potential and a second power supply potential, the internal circuit comprising an input terminal coupled to receive an input signal and an output terminal coupled to provide an output signal, the internal circuit comprising a first insulated gate field effect transistor (IGFET) formed in the second region, the first IGFET comprising a plurality of channels that extend in a first direction and are substantially aligned in a second direction to that crosses the first direction, and a control gate that substantially surrounds the plurality of channels, wherein the plurality of channels provide a controllable impedance path between a first source/drain terminal and a second source/drain terminal; and a first electrostatic discharge (ESD) protection circuit structure that is provided in the first region, and is electrically connected between the first power supply potential and the second power supply potential, wherein the control gate is coupled to receive the input signal and the first source/drain terminal is coupled to provide the output signal, and the internal circuit is not connected to receive a signal generated external to the integrated circuit device and is not connected to provide a signal external to the integrated circuit device.
2. The integrated circuit device of claim 1, wherein: the first ESD protection circuit structure comprises a diode.
3. The integrated circuit device of claim 2, wherein: the diode comprises an anode terminal

electrically connected to the first power supply potential and a cathode region electrically connected to the second power supply potential.

4. The integrated circuit device of claim 1, wherein: the first ESD protection circuit structure comprises a silicon controlled rectifier (SCR) circuit coupled between the first power supply potential and the second power supply potential.

5. The integrated circuit device of claim 1, wherein: the internal circuit further comprises a second IGFET formed in the second region, the second IGFET comprising a plurality of channels that extend in the first direction and are substantially aligned in the second direction, and a control gate that substantially surrounds the plurality of channels, wherein the plurality of channels provide a controllable impedance path between a third source/drain terminal and a fourth source/drain terminal, and the third source/drain terminal is coupled to the first source/drain terminal; and the first IGFET is a p-type IGFET and the second IGFET is a n-type IGFET.

6. The integrated circuit device of claim 5, wherein: the p-type IGFET and n-type IGFET are configured as an inverter circuit.

7. The integrated circuit device of claim 1, wherein: the first and second power supply potentials are provided from a source that is external to the integrated circuit device.

8. The integrated circuit device of claim 1, further comprising: an input buffer circuit coupled to receive an input signal provided from a source that is external to the integrated circuit device, the input signal received at a first pad.

9. The integrated circuit device of claim 8, further comprising: a second ESD protection circuit structure having a first terminal electrically connected between the first pad and an input terminal of the input buffer circuit.

10. The integrated circuit device of claim 9, wherein: the second ESD protection circuit structure comprises a resistor electrically connected between the first terminal of the second ESD protection circuit structure and the input terminal of the input buffer circuit.

11. The integrated circuit device of claim 10, wherein: the second ESD protection circuit structure further comprises a first diode having an anode terminal coupled to the first terminal of the second ESD protection circuit structure.

12. The integrated circuit device of claim 11, wherein: the second ESD protection circuit structure further comprises a second diode having a cathode terminal coupled to the first terminal of the second ESD protection circuit structure.

13. The integrated circuit device of claim 11, wherein: the first diode comprises a plurality of current carrying regions disposed substantially in the first direction.

14. The integrated circuit device of claim 1, further comprising: an output buffer circuit coupled to receive an input signal and provide an output signal externally to the integrated circuit device through a first pad.

15. The integrated circuit device of claim 14, further comprising: a second ESD protection circuit structure having a first terminal electrically connected between the first pad and an output terminal of the output buffer circuit.

16. The integrated circuit device of claim 15, wherein: the second ESD protection circuit structure further comprises a first diode having an anode terminal coupled to the first terminal of the second ESD protection circuit structure.

17. The integrated circuit device of claim 16, wherein: the second ESD protection circuit structure further comprises a second diode having a cathode terminal coupled to the first terminal of the second ESD protection circuit structure.

18. The integrated circuit device of claim 16, wherein: the first diode comprises a plurality of current carrying regions disposed substantially in the first direction.

19. The integrated circuit device of claim 1, wherein: the first ESD protection circuit structure is provided in a semiconductor substrate.

20. An integrated circuit device, comprising: a first region; a second region over the first region; an

internal circuit coupled between a first power supply potential and a second power supply potential, the internal circuit comprising an input terminal configured to receive an input signal, an output terminal configured to provide an output signal, and a first insulated gate field effect transistor (IGFET), wherein the first IGFET is provided in the second region, comprises a plurality of channels and a control gate configured to receive the input signal, and is configured to provide a controllable impedance path between a first source/drain terminal and a second source/drain terminal; and a first electrostatic discharge (ESD) protection circuit structure that is provided in the first region, and is electrically connected between the first power supply potential and the second power supply potential.
